

Optimization report

Part 1

	Before index	After index
Documents Scanned	33	2
Scan strategy	COLLSCAN	IXSCAN
Execution time	0 ms	1 ms

Part 2

1. Why indexes improve performance.
They create an order in data and that allows to find the right one documents faster (because we know where are they).
2. Differences between COLLSCAN and IXSCAN.
COLLSCAN – performs whole collection scan, reading and processing every document.
IXSCAN – perform scan based on index, so doesn't scan whole collection, but only relevant data.
3. How many documents were examined before and after indexing.
Before index all 33 documents were examined, but after index creation only 2 were examined.
4. Whether execution time improved.
From execution plans, it hasn't improved, but got slower. But it is wrong to judge on such small database. If it was big dataset – time would have been improved.

Screenshots

Before index:

```
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookstore.books',
    parsedQuery: {
      $and: [
        {
          category: {
            $eq: 'Urban Fantasy'
          }
        },
        {
          published_year: {
            $gte: NumberInt('2021')
          }
        }
      ]
    },
    indexFilterSet: false,
    queryHash: '92EDD004',
    planCacheShapeHash: '92EDD004',
    planCacheKey: '195A970C',
    optimizationTimeMillis: NumberInt('0'),
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: {
        $and: [
          {
            category: {
              $eq: 'Urban Fantasy'
            }
          }
        ]
      }
    }
  }
}
```

```
{
  published_year: {
    $gte: NumberInt('2021')
  }
}
],
  },
  direction: 'forward'
},
rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: NumberInt('2'),
  executionTimeMillis: NumberInt('0'),
  totalKeysExamined: NumberInt('0'),
  totalDocsExamined: NumberInt('33'),
  executionStages: {
    isCached: false,
    stage: 'COLLSCAN',
    filter: {
      $and: [
        {
          category: {
            $eq: 'Urban Fantasy'
          }
        },
        {
          published_year: {
            $gte: NumberInt('2021')
          }
        }
      ]
    }
  },
  nReturned: NumberInt('2'),
  executionTimeMillisEstimate: NumberInt('0'),
}
```

After index:

```
explainVersion: '1',
queryPlanner: {
  namespace: 'bookstore.books',
  parsedQuery: {
    $and: [
      {
        category: {
          $eq: 'Urban Fantasy'
        }
      },
      {
        published_year: {
          $gte: NumberInt('2021')
        }
      }
    ]
  },
  indexFilterSet: false,
  queryHash: '92EDD004',
  planCacheShapeHash: '92EDD004',
  planCacheKey: '795A952C',
  optimizationTimeMillis: NumberInt('0'),
  maxIndexedOrSolutionsReached: false,
  maxIndexedAndSolutionsReached: false,
  maxScansToExplodeReached: false,
  prunedSimilarIndexes: false,
  winningPlan: {
    isCached: false,
    stage: 'FETCH',
    inputStage: {
      stage: 'IXSCAN',
      keyPattern: {
        category: NumberInt('1'),
        published_year: NumberInt('1')
      },

```

```
isMultiKey: false,
multiKeyPaths: {
  category: [],
  published_year: []
},
isUnique: false,
isSparse: false,
isPartial: false,
indexVersion: NumberInt('2'),
direction: 'forward',
indexBounds: {
  category: [
    ["Urban Fantasy", "Urban Fantasy"]
  ],
  published_year: [
    [2021, inf.0]
  ]
}
},
rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: NumberInt('2'),
  executionTimeMillis: NumberInt('1'),
  totalKeysExamined: NumberInt('2'),
  totalDocsExamined: NumberInt('2'),
  executionStages: {
    isCached: false,
    stage: 'FETCH',
    nReturned: NumberInt('2'),
    executionTimeMillisEstimate: NumberInt('0'),
    works: NumberInt('3'),
    advanced: NumberInt('2'),
    needTime: NumberInt('0'),

```